DATA SCIENTIST INTERVIEW

Use Case Assessment

MediCare Patient Follow-Up Agent

Domain	Role	Format	Time Limit
Healthcare	Data Scientist	Jupyter Notebook + CSV	40 Minutes

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1. Scenario Background

MediCare Clinic is a mid-sized outpatient healthcare facility managing over 1,000 chronic disease patients. The clinical team struggles to proactively manage follow-up care across patients with conditions such as Type 2 Diabetes, Hypertension, Heart Failure, COPD, and Anemia. Patients who miss appointments or show worsening lab values are often not identified until their next scheduled visit, leading to avoidable complications and hospital readmissions.

The clinic's medical director wants to deploy an AI-powered agent that can autonomously review patient records, identify those at risk, and generate prioritised action plans for the care coordination team.

Your role:

You are a Data Scientist hired to build a prototype of this agent using Agentic AI. You will be given a dataset of 100 patients and must build an AI agent that autonomously orchestrates tool calls to analyse patient data, flag clinical risks, and generate follow-up care actions.

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2. Dataset Description

The file patient_data.csv contains 100 anonymised patient records. Each row represents one patient's most recent clinical snapshot.

Column	Type	Description
patient_id	String	Unique patient identifier (e.g., P0001)
patient_name	String	Patient full name
age	Integer	Patient age in years
gender	String	Male / Female
diagnosis	String	Primary diagnosis (may be compound)
current_medication	String	Current prescribed medication(s)
lab_test	String	Most recent lab test name
lab_value	Float	Numerical result of the lab test
lab_unit	String	Unit of measurement (%, mmHg, g/dL, etc.)
last_visit_date	Date	Date of the patient's last clinic visit

next_scheduled_visit	Date	Date of next scheduled appointment
days_since_last_visit	Integer	Days elapsed since last visit
missed_last_appointment	String	Yes / No — did patient miss their last appointment
vitals_bp_systolic	Integer	Systolic blood pressure (mmHg)
vitals_bp_diastolic	Integer	Diastolic blood pressure (mmHg)
vitals_heart_rate	Integer	Heart rate (beats per minute)
vitals_spo2	Integer	Blood oxygen saturation (%)
notes	String	Free-text clinical notes from last visit

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3. Tasks

Complete all tasks in a single Jupyter Notebook. Your notebook must be self-contained and runnable. The agent must use an LLM (e.g., Grok, OpenAI, Anthropic API etc) to make decisions — do not hardcode logic for actions that should be decided by the LLM.

Task 1 — Data Loading & Exploratory Analysis

Task 2 — Define Agent Tools

Task 3 — Implement the Agentic Loop

Task 4 — Single Patient Analysis

Task 5 — Missed Appointment Follow-Up

5. Submission Requirements

- A single Jupyter Notebook (.ipynb) with all cells run and outputs visible.
- The patient_data.csv file placed in the same directory as the notebook.
- The notebook must run end-to-end without errors when rerun from top.
- Use the LLM API (You can mask the sAPI Key)
- Total code (excluding markdown and comments) must not exceed 250 lines.
- Submit via the link shared in your interview invitation email before the scheduled interview date.

Good luck! We look forward to discussing your implementation.